F-TFM: Accelerating Total Focusing Method for Ultrasonic Array Imaging on FPGA

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***Abstract*—The total focusing method (TFM) is a widely employed imaging algorithm utilized for processing ultrasonic signals in various applications. Currently, there is a growing demand for real-time efficient TFM processing. However, existing research struggles to achieve satisfactory throughputs and optimal energy efficiency. In this work, we present F-TFM, an efficient TFM imaging system on FPGA. Firstly, we analyze and explore the trade-off between re-computation and memory access in TFM and organize the efficient workflow for high-efficient processing on FPGA. Secondly, we propose an efficient accelerator architecture with carefully designed dataflow as well as optimized hardware parameter configurations. Evaluations demonstrate that F-TFM achieves remarkable improvements over existing works, while maintaining high image quality. Specifically, F-TFM achieves up to $5.43\times$ speedups and $47.15\times$ energy efficiency improvements compared with the high-end GPU baseline and achieves up to $35.94\times$ speedups compared with the FPGA baselines.**

***Index Terms*—FPGA Acceleration, High-level Synthesis (HLS), Total Focusing Method (TFM)**

I. Introduction

The total focusing method (TFM) is widely used in ultrasound imaging for applications such as material testing, non-destructive testing, and medical imaging [1]. Recently, there is an increasing demand for efficient real-time or high-frame-rate TFM-based imaging, which requires high-performance and energy-efficient computing platforms due to the computational and memory requirements in the TFM algorithm.

There have been several existing works on the TFM acceleration using heterogeneous computing platforms. However, they either focused on computational kernel optimizations only or lacks modern SW/HW co-design methodologies, struggling to achieve an satisfactory balance between throughput, energy efficiency, and imaging quality. In this paper, we introduces F-TFM, an FPGA-based TFM imaging system. The distinguishing features of F-TFM from the related works are:

- We analyze the trade-off between computation and memory access in the TFM algorithm and optimize the workflow for efficient TFM processing on FPGA.
- We propose a novel FPGA-based accelerator architecture for TFM with optimized dataflow organization as well as hardware parameter configurations.
- Experiments demonstrate that F-TFM achieves up to $5.43\times$ speedups and $47.15\times$ energy efficiency improvements compared to the GPU baselines and achieves up to $35.94\times$ speedups compared with FPGA baselines.

II. Background

TFM Algorithm and Workflow: TFM imaging system uses an 1D phased array ultrasonic probe with K transceivers to emit and receive ultrasonic waves. The system employs full matrix capture (FMC) to capture signals by using all/multiple possible transceiver combinations ($\langle i,j\rangle, 0 \le i,j < K$). The inspected surface in the X-Y direction is discretized into an $M \times N$ matrix $I(P)$, and the time delay computation (DLC) between the transceiver pair $\langle i,j\rangle$ and P is calculated as:

$$T_{iP} + T_{Pj} = \frac{\sqrt{(x_i-x)^2+(z_i-z)^2}+\sqrt{(x_j-x)^2+(z_j-z)^2}}{c}$$

where c denote the velocity of ultrasonic waves. The values of DLC are crucial for beamforming and focusing the ultrasonic waves to generate the final image $I[P(x,z)]$:

$$I[P(x,z)] = |\sum_i \sum_j h_{ij}(T_{iP} + T_{Pj})| \tag{1}$$

where h_{ij} represents the data preprocessing (DPP) method using Hilbert transform, and pixel interpolation (PIT) methods such as linear or nearest-neighbor methods are then applied.

Initially, with the basic configurations and relevant statistical information of the ultrasonic data, the DLC can be performed in advance. Then, the DPP and PIT functions are executed frame by frame to enable continuous imaging.

Related Works: Several studies have explored accelerating approaches on GPUs and FPGAs. Romero et al. [2] analyzed parallelization strategies on GPUs. The Bristol UNDT Group developed Brain [3], an open-source GPU-accelerated TFM imaging software. They mainly focused on kernel optimization and neglected the data transfer overhead, leading to an unsatisfactory throughput and energy-efficiency. M. Nijiki et al. [4] proposed a multi-FPGA system for real-time TFM reconstruction. But the lack of SW/HW co-design limited the system flexibility and scalability. C. Wang et al. [5] addressed the computation complexity by adopting pixel-wise Hilbert transform, which is oversimplified and leading to unexpected outcomes. In F-TFM, we adopt SW/HW co-design methodologies in the FPGA-based accelerator design to achieve a better balance between imaging quality, throughput, and power.

III. F-TFM System

Software Analysis and Workflow Organization: Consider a TFM-based imaging system that consists of a phased array with K transceivers and operates over T time-steps, generating

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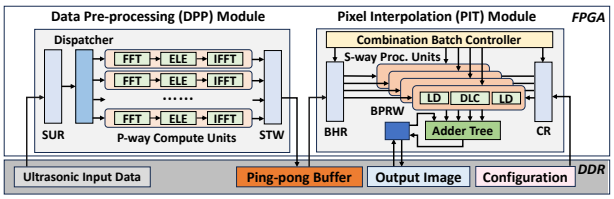

Fig. 1. F-TFM architecture. SUR, STW, BHR, CR, and BPRW denote streaming ultrasonic data reader, streaming transposed data writer, batched Hilbert-transformed data reader, configuration reader, and batched pixel reader/writer.

an $M \times N$ image. According to the TFM principles, we can find that two main features: a) the data reuse in the pre-calculated DLC results will cause large overhead (size: $O(MNK^2)$), which should be optimized by reducing memory access volume; b) the DLC, DPP, and PIT stages are relatively decoupled, which should be organized in an efficient workflow to reduce the context switching overhead.

Therefore, we first propose the **real-time pixel-wise DLC** strategy by calculating pixel-wise delays repeatedly when processing the corresponding pixels at every frame. This trade-off between recomputation and data reuse makes F-TFM reduce more than 90% memory access in the data loading of the PIT stage, at the cost of parallel pixel-wise DLC logic for real-time computation. Secondly we propose a dataflow architecture in F-TFM with **pipelined DPP and PIT execution**, where the pixel-wise DLC can be integrated in the PIT module. The dataflow effectively utilizes the computation resources on FPGA and significantly improves throughput. Besides, F-TFM can also optimize the throughput and resource utilization holistically by parallelisms settings.

Hardware Accelerator Design: The architecture of F-TFM accelerator is demonstrated on Figure 1. The basic configurations are preloaded on-chip. The accelerator reads the FMC data and generates the images continuously. Considering the large Hilbert-transformed data (size: $O(TK^2)$) and the orthogonal memory access patterns between DPP and PIT, there is an in-DDR ping-pong buffer for efficient data transfer without the over-utilization of BRAMs or URAMs.

In the DPP module, the Hilbert transform is performed through row-wise FFT/IFFT. Given the FMC data (size: $O(TK^2)$), the scheduler dispatches T tasks to P-way compute units (CUs), while STW writes the results in a transposed form to DDR for further efficient loading in PIT. In the PIT module, pixel processing is performed in batch among the $O(K^2)$ transceiver pairs. S-way processing units (PUs) are employed for pixel-wise DLC, data extraction, and amplitude calculation. And the adder tree is for efficient value accumulation.

IV. Evaluations

Setup: The F-TFM prototype is assessed on Xilinx Kintex UltraScale+ XCKU19P using Vitis HLS v2022.2 and synthesized at 200MHz. The benchmarks (selected from Brain), hardware configurations, and resource utilization are in Tab. I, where D denotes the 3D image depth. The datatype in F-TFM is `ap_fixed<32,16>` for image quality. For comparisons, we evaluate the implementations in Brain on NVIDIA RTX 3080Ti and Jetson TX1. And we calculate the corresponding performance with the results reported in [4] and [5].

TABLE I
SW/HW Configurations and Resource Utilization on XCKU19P

Case	SW/HW Config							Resource Utilization(%)					
	T	K	M	N	D^*	P	S	BRAM	DSP	FF	LUT	URAM	BW
alu-1d	1000	64	394	249	1	12	128	22	49	15	30	18	59
com-1d	500	64	410	442	1	16	256	28	73	15	31	22	45
obl-1d	1400	64	1123	971	1	10	256	22	66	15	30	14	45
imm-1d	1001	64	1073	331	1	16	256	28	73	15	31	22	55
alu-2d	500	128	15	15	32	14	64	72	38	14	26	80	32
ani-2d	600	128	20	20	33	14	64	72	38	14	26	80	35

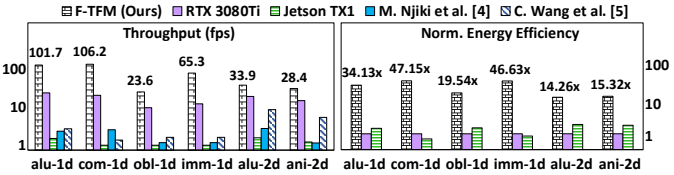

Fig. 2. Throughput and normalized energy efficiency comparisons.

Results and Analysis: Fig. 2 shows the throughput and normalized energy efficiency comparisons between F-TFM and all baselines. For throughputs, F-TFM achieves high-frame-rate imaging (up to 106.2fps) and significant speedups, which are ranging from $1.83\times$ to $5.43\times$ compared to NVIDIA RTX 3080Ti. As for the FPGA baseline [5], the speedups are ranging from $3.76\times$ to $35.94\times$. We also compare the normalized frame rates per Watt (fps/W) between F-TFM and only GPU baselines due to the lack of published power data for the FPGA baselines. Leveraging its specialized architecture and well-organized dataflow, F-TFM achieves remarkable efficiency gains over GPUs, ranging from $14.26\times$ to $47.15\times$. For resource utilization, we observe high DSP utilization ($\geq$49%) when K=64 and high BRAM/URAM utilization when K=128. Besides, in `alu-2d` and `ani-2d`, the performance is limited by the memory bandwidth in STW.

V. Conclusion

In this paper, we introduce F-TFM, an efficient TFM imaging system on FPGA. With the analysis of computation and memory access trade-off, we propose an efficient workflow and design an accelerator architecture with carefully designed dataflow and well-optimized parameters. Evaluation shows that F-TFM achieves high-performance, energy-efficient processing with quality reconstruction.

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